LOVELY PROFESSIONAL UNIVERSITY

Sample Question Set - Unit 6

Course Code: CSE252	Unit: 6
Course Title: INTRODUCTION TO ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING	Unit Title: AI for Intelligent Systems and Robotics
Course Outcome: CO6	Unit Mapped RBT: L3 (Apply)
Paper: Sample Subjective Questions	Total Questions: 10
Maximum Marks: 10 x 10 = 100	Suggested Duration: Not timed - practice set

Subjective Questions (10 x 10 = 100 Marks)

Q1. Apply reinforcement learning to a robot vacuum cleaner and describe the exploration problem it faces. (10 Marks)

(a) State what the agent, the state, the action and the reward are in this task.

(b) Use the exploration and exploitation exchange to explain what goes wrong at each of the two extremes.

[L3 - Apply]

Q2. Compare a rule-based chatbot with one built on a learned language model. (10 Marks)

(a) State how each of the two decides what a user's message means.

(b) Use predictability, coverage and auditability to select one for a university helpdesk, with reasons.

[L3 - Apply]

Q3. Examine a robot vision model that performs well in testing and then fails once it is in service. (10 Marks)

(a) State two distinct explanations for the gap between the two settings.

(b) Use one of the two to describe the data you would gather and the check you would add.

[L3 - Apply]

Q4. Use the steps of natural language processing to prepare a user's message for a chatbot. (10 Marks)

(a) Name the steps in order and state what each one produces.

(b) Use a sample message to show what the system extracts before it can act on the request.

[L3 - Apply]

Q5. Apply computer vision to a quality inspection task on a production line. (10 Marks)

(a) State whether classification or detection is required, and what the model receives and returns.

(b) Use the cost of the two kinds of error to describe where you would place the confidence threshold.

[L3 - Apply]

Q6. Compare running a vision model on board a robot with running it on a remote server. (10 Marks)

(a) State how the two compare on latency, on connectivity and on the computing power available.

(b) Use a robot that must stop for obstacles to select one of the two, and state what is given up.

[L3 - Apply]

Q7. Examine a reward function that produces behaviour nobody intended. (10 Marks)

(a) State, with an example, how a reward measured over the agent's activity can be satisfied without the task being done.

(b) Use this to describe how the reward should have been written, and what you would monitor after deployment.

[L3 - Apply]

Q8. Use transfer learning to build a vision model from a small collection of labelled images. (10 Marks)

(a) State what is reused from the pre-trained network and what is trained afresh.

(b) Use the size of the labelled set to explain one problem that can arise and one way of reducing it.

[L3 - Apply]

Q9. Examine the safety requirements of an autonomous robot that works alongside people. (10 Marks)

(a) State two failures that would be dangerous and what each one would result from.

(b) Use those failures to describe the safeguards you would build in before deployment.

[L3 - Apply]

Q10. Relate the future trends of Artificial Intelligence to the design of autonomous robots. (10 Marks)

(a) State two trends and describe what each one would change about how a robot is built.

(b) Use one of the two to explain a benefit it brings and a new risk it introduces.

[L3 - Apply]

----- End of Sample Question Set -----